

SENG 326 Software Architecture Group Project

Task 2 Architectural Refactoring of Checkstyle 13.2.0

Part A: CheckStyle Refactoring

In this Task you will refactor an architectural slice of Checkstyle 13.2.0 into your group's target architecture. This means that only a subset of Checkstyle will be refactored into the new architecture and the rest of the application will remain unchanged. Your refactored application must execute correctly and function exactly like the original Checkstyle.

Architectural Slice Scope

You will refactor all the checks belonging to the Metrics and Size Violations categories.

This means:

- These checks must be executed exclusively through your refactored target architecture.
- All other Checkstyle checks and architecture must remain unchanged.
- No hard-coding or workarounds
- Existing configuration mechanisms must work
- Check logic must not be rewritten

Constraints

- The target architecture must be fully implemented within the slice
- No architectural “shortcuts” into other code from inside the slice
- You may add new classes in order to implement the target architecture:
 - Only add the minimal number of classes necessary
 - Every new class must have:
 - A clear architectural role and justification
 - A single responsibility

After Refactoring:

- Checkstyle must execute correctly and give correct output as the original version:
 - You must show an execution of pre and post refactoring using a file with multiple violations including all metrics and size violations.
 - We must be able to execute and test the post refactoring version with NO amendments.
- You must “prove” the implementation of the target architecture using the tools and steps in Task 1.

Report A: In addition to the above, the report should contain the following:

1- Target Architectural Style and Architectural Slice

Clearly define and provide diagrams (C4, UML, dependency diagrams, etc) showing:

- The Checkstyle components that were included in the architectural slice and those that were excluded.
- The boundary between the refactored slice and the remaining system
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Part B: Scalability and Performance Evaluation

Repeat the same experiments from Task 1 Part B for your refactored application. **Use the same experimental setup and input Java codebases.**

Compare the results with those in Task 1 Part B. Plot both sets of results on one graph.

Analyze the results of the refactored application in comparison to the original application in terms of:

- performance and scalability.
- Impact of refactoring
- the bottlenecks, etc.

Report Part B: Your report should contain all the above and the following:

- The analysis of the results with respect to the original and refactored architectures
- What would you do differently
- Lessons Learned

Deadline for Task 2: Saturday May 16th 12 midnight.

Submission:

Only ONE student should submit for each group.

Submit: a ZIP file containing two files

(1) the final report in non-protected pdf file and

(2) java source code file for the application [The source code should clearly show where the refactoring has been implemented].

name the zip file using the group's name.